Probability and Statistics

Chapter 2: Probability

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Chapter 2. Probability

Sample Spaces and Events

A **random experiment (phép thử ngẫu nhiên hoặc thí nghiệm)** is any activity or process whose outcome is subject to uncertainty.

Definition

The sample space of an experiment **(không gian mẫu)**, denoted by S, is the set of all possible outcomes of that experiment.

For other materials, the sample space of an experiment is denoted by Ω.

Example

- If the experiment consists of flipping a coin, then the sample space consists of the following two points

$$S = \{H(head), T(tail)\}.$$

- When a six-sided die is rolled, the sample space

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Definition

An event (sự kiện hoặc biến cố) is any collection (subset) of outcomes contained in the sample space S. An event is simple if it consists of exactly one outcome and compound if it consists of more than one outcome.

An event is just a set, so relationships and results from elementary set theory can be used to study events. The following operations will be used to create new events from given events.

Definition

1. The complement of an event A, denoted by A', is the set of all outcomes in S that are not contained in A (Note that $A' = \overline{A} = S \setminus A$).
2. The union of two events A and B, denoted by $A \cup B$ and read "A or B", is the event consisting of all outcomes that are either in A or in B or in both events (so that the union includes outcomes for which both A and B occur as well as outcomes for which exactly one occurs)-that is, all outcomes in at least one of the events (or $A \cup B = A + B$).
3. The intersection of two events A and B, denoted by $A \cap B$ and read "A and B", is the event consisting of all outcomes that are in both A and B (or $A \cap B = AB$).

Definition

Let $\emptyset$ denote the null event (the event consisting of no outcomes whatsoever). When $A \cap B = \emptyset$, A and B are said to be mutually exclusive or disjoint events (hai sự kiện xung khắc).

Example

1. Suppose that a dice is rolled and the upturned faces will equal to a prime number. What is the A'?
2. Suppose that a dice is rolled and $A_1 = \{1\}$, $A_2 = \{2\}$. Clearly, A_1 and A_2 are two disjoint events (other words, $A_1 \cap A_2 = \emptyset$),

Axioms, Properties

Given an experiment and a sample space S, the objective of probability is to assign to each event A a number $P(A)$, called the probability of the event A, which will give a precise measure of the chance that A will occur.

Properties

1. For any event A, $0 \leq P(A) \leq 1$.
2. $P(\emptyset) = 0$.
3. $P(S) = 1$.
4. $P(A') = 1 - P(A)$.
5. If $A_1, A_2, A_3, \ldots$ is an infinite collection of disjoint events, then $P(A_1 \cup A_2 \cup \ldots) = \sum_i^{\infty} P(A_i)$.

When the various outcomes of an experiment are equally likely (the same probability is assigned to each simple event), the task of computing probabilities reduces to counting.
Letting N denote the number of outcomes in a sample space and $N(A)$ represent the number of outcomes contained in an event A,

$$P(A) = \frac{N(A)}{N}.$$

Example

Suppose that a dice is rolled and the upturned faces will equal a prime number.

a. What is the probability $P(A)$?
b. What is the probability $P(A')$?

Properties

1. For any two events A and B

$$P(A \cup B) = P(A + B) = P(A) + P(B) - P(A \cap B).$$

2. For any three events A, B, and C,

$$\begin{aligned} P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) \\ -P(A \cap C) - P(B \cap C) + P(A \cap B \cap C). \end{aligned}$$

Note that, $P(A \cup B) = P(A + B)$, $P(A \cap B) = P(AB)$.

Example

1. (a) Let A, B be two events in a random experiment, where $P(A) = 0{,}45$, $P(B) = 0{,}3$ và $P(A \cap B) = 0{,}1$. Determine $P(A \cup B)$, $P(A \setminus B)$ và $P(B \setminus A)$.
2. (b) Let A, B, C be three mutually exclusive events in a random experiment, where $P(A) = 0, 1$, $P(B) = 0{,}3$ và $P(C) = 0{,}4$. Compute $P(A' \cap B' \cap C')$.

Remark

Let A, B and C be events in a random experiment, then

- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$;
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
- $(A \cup B)' = A' \cap B'$;
- $(A \cap B)' = A' \cup B'$;

Conditional Probability

In this section, we examine how the information "an event B has occurred" affects the probability assigned to A. We will use the notation $P(A|B)$ to represent the conditional probability of A given that the event B has occurred; B is the "conditioning event".

Definition

For any two events A and B with $P(B) > 0$, the conditional probability of A given that B has occurred is defined by

$$P(A|B) = \frac{P(AB)}{P(B)}.$$

Example

A coin is flipped twice. Assuming that all four points in the sample space

$$S = (H, H), (H, T), (T, H), (T, T)$$

are equally likely, what is the conditional probability that both flips land on heads, given that

- (a) the first flip lands on heads?
- (b) at least one flip lands on heads?

Corollary

$$P(A \cap B) = P(AB) = P(B)P(A|B) = P(A)P(B|A).$$

Example

Let A and B be two random events that satisfy $P(A) = 0{,}5$, $P(B) = 0{,}6$ and $P(A \cap B) = 0,2$. Determine $P(A|B)$?

Example

Suppose that a dice is rolled. Let A be the event that the upturned faces equal a prime number and B be the event that the surface appears as 2. Compute $P(B|A)$?

Recall that events $A_1, \ldots, A_n$ are mutually exclusive if no two have any common outcomes. The events are exhaustive if one A_i must occur, so that $A_1 \cup A_2 \cdots \cup A_n = \Omega = S$. (**Nhóm các sự kiện đầy đủ**).

Theorem (The Law of total Probability)

Let $A_1, \ldots, A_n$ be mutually exclusive and exhaustive events. Then for any other event B,

$$\begin{aligned} P(B) = & \sum_{i=1}^{n} P(A_i)P(B|A_i) \\ = & P(A_1)P(B|A_1) + \ldots P(A_n)P(B|A_n) \end{aligned}$$

Example

An insurance company believes that people can be divided into two classes: those who are accident prone and those who are not. The company's statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability 0,4, whereas this probability decreases to 0,2 for a person who is not accident prone. Assume that 30 percent of the population is accident prone.

- what is the probability that a new policyholder will have an accident within a year of purchasing a policy?

Hint: Let A denote the event that the policyholder is accident prone.
$\Rightarrow A'$ is the event that the policyholder is not accident prone.
Let B denote the event that the policyholder will have an accident within a year of purchasing the policy.
By using the Law of total Probability, we now have

$$\begin{aligned} P(B) =& P(A)P(B|A) + P(A')P(B|A') \\ =& 0{,}3 \times 0{,}4 + 0{,}7 \times 0{,}2 = 0{,}26. \end{aligned}$$

Theorem(Bayes)

Let $A_1, A_2, \ldots, A_n$ be a collection of n mutually exclusive and exhaustive events with prior probabilities $P(A_i)$ $(i = 1, \ldots, n)$. Then for any other event B for which $\mathrm{P}(B) > 0$, the posterior probability of A_k given that B has occurred is

$$P(A_k|B) = \frac{P(A_k \cap B)}{P(B)} = \frac{P(B|A_k)P(A_k)}{\sum\limits_{i=1}^{n} P(B|A_i)P(A_i)},\ k = 1, \ldots, n$$

Example

Suppose that a new policyholder has an accident within a year of purchasing a policy. What is the probability that he or she is accident prone?

Example

A bin contains 3 different types of disposable flashlights. The probability that a type-1 flashlight will give over 100 hours of use is 0,7, with the corresponding probabilities for type-2 and type-3 flashlights being 0,4 and 0,3, respectively. Suppose that 20 percent of the flashlights in the bin are type-1, 30 percent are type-2, and 50 percent are type-3.

a. What is the probability that a randomly chosen flashlight will give more than 100 hours of use?
b. Given that a flashlight lasted over 100 hours, what is the conditional probability that it was a type-2 flashlight?

Independence

Definition

Two events are called **independent (độc lập)** if the occurrence of one event does not change the probability of the other event. Equivalently, any one of the following equivalent statements is true

- $P(A|B) = P(A)$;
- $P(A \cap B) = P(A)P(B)$;
- $P(B|A) = P(B)$.

Remark: If A and B are independent events then so are events A and B', events A' and B, and events A' and B'.

Example

Two coins are flipped, and all 4 outcomes are assumed to be equally likely. If E is the event that the first coin lands on heads and F the event that the second lands on tails, then E and F are independent?

Since $P(EF) = P((H,T)) = \frac{1}{4}$, whereas $P(E) = P((H,H),(H,T)) = \frac{1}{2}$ and $P(F) = P((H,T),(T,T)) = \frac{1}{2}$.

Example

Two fair dice are rolled. What is the conditional probability that at least one lands on 6 given that the dice land on different numbers?